

Healthcare Capability Intelligence Pipeline using Databricks



Team Name:

KRIPA



Program:

**Databricks Accenture Challenge
Hackathon 2026**



Theme:

Data Engineering



KIIT University



Problem Statement

Healthcare data is available in unstructured format which makes it difficult to analyze treatment capabilities, costs and patient care services. Manual analysis is time-consuming and error-prone.



Objective

The objective of this project is to build an end-to-end data analytics pipeline that:

- ✓ Extracts key capabilities from healthcare data using Intelligent Document Parsing (IDP)
- ✓ Cleans and preprocesses the dataset
- ✓ Stores the processed data in Delta Lake
- ✓ Performs SQL based analytics
- ✓ Visualizes insights using Power BI

Data Pipeline Workflow



Implementation Details

- ✓ **Data Ingestion:** Loaded dataset from Kaggle using PySpark.
- ✓ **Data Cleaning:** Handled missing values and created structured dataset.
- ✓ **IDP (Intelligent Parsing):** Extracted healthcare capabilities from text using rule-based detection.
- ✓ **Storage:** Saved processed data in Delta Lake.
- ✓ **Analytics:** Performed SQL queries for analysis.
- ✓ **Visualization:** Connected Power BI for interactive dashboard.

TECH STACK & TASK MAPPING

Tech Stack



Python
(PySpark)



Databricks
(Notebook, SQL)



Delta Lake
(Storage)



SQL
Warehouse



Power BI
(Visualization)

Task Mapping (As per Project Helper)



Task 1: Python Basics

Data loading, cleaning and preprocessing using PySpark



Task 2: Advanced Python

Intelligent Document Parsing (IDP) using custom rules



Task 3: Advanced SQL

SQL queries for analytics and insights



Task 4: Data Ingestion

Ingesting data from CSV to Delta Lake



Task 5: Final Project

End-to-end pipeline: ingestion → processing → storage → analytics → visualization



Note: Databricks components used in this project are Notebooks, Delta Lake and SQL Warehouse.

Key Insights



Cardiology and General Treatment are the most common capabilities.



Surgery related treatments have higher average cost.



Emergency Care plays a critical role in patient support.



The pipeline provides clear insights for decision making.

Dashboard Preview (Power BI)

Healthcare Capability Intelligence Dashboard

Databricks + Intelligent Document Parsing Analytics

Total Treatments

200

Total Healthcare Cost

551.25K

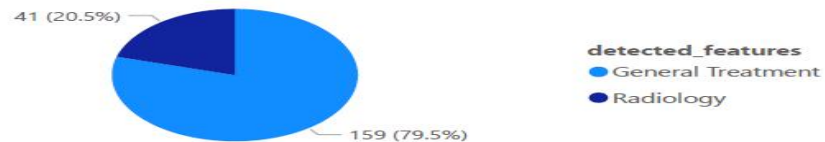
Detected Features

2

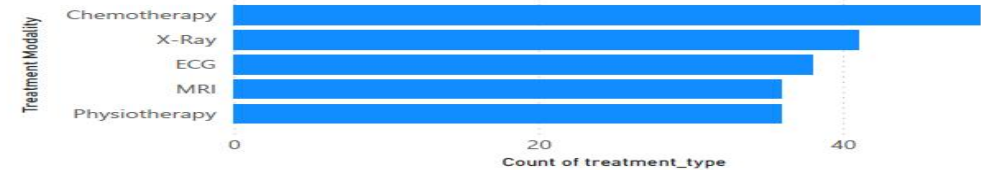
Average Treatment Cost

2.76K

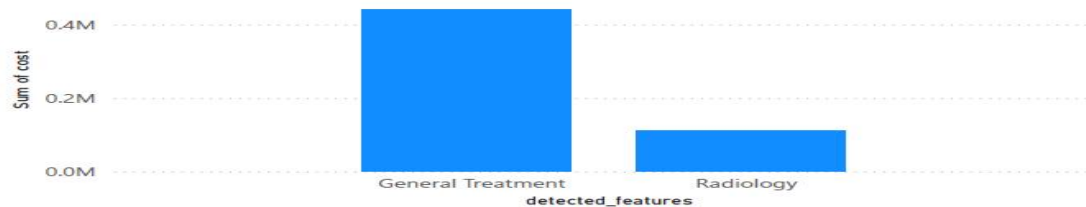
Intelligent Document Parsing (IDP) Insights



Treatment Modality Distribution



Financial Impact by Extracted Health Feature



Patient Treatment Records & Documentation

treatment_type	Sum of cost	description	detected_features
Chemotherapy	33,854.13	Advanced protocol	General Treatment
Chemotherapy	34,395.88	Basic screening	General Treatment
Chemotherapy	60,605.67	Standard procedure	General Treatment
ECG	53,558.78	Advanced protocol	General Treatment
ECG	20,790.71	Basic screening	General Treatment
ECG	21,874.75	Standard procedure	General Treatment
MRI	30,277.62	Advanced protocol	General Treatment
MRI	55,394.68	Basic screening	General Treatment
Total	5,51,249.85		

Figure 1: Healthcare Analytics Dashboard created using Power BI



Conclusion

This project demonstrates an end-to-end Healthcare Capability Intelligence Pipeline using Databricks. It efficiently extracts capabilities from unstructured data, stores it in Delta Lake, performs SQL analytics and visualizes insights using Power BI. This solution can support hospitals and healthcare organizations in making data-driven decisions.



Future Improvements



Implement ML models for cost prediction and trend analysis



Use NLP for advanced text understanding



Build real-time streaming pipeline



Deploy pipeline on cloud for scalability



Enhance dashboard with more KPIs and filters

